

Aspect .spc Spectrum File Format Verification Report

Comparison of "Format of spectrum file.h" against the provided Aspect .spc sample archive

Technical English version | Generated from supplied files in /mnt/data

Executive summary

The provided header file matches the supplied Aspect .spc sample files. The binary file consists of a 512-byte packed tSpectrHeader followed by raw spectrum counts stored as DWORD / uint32 little-endian values.

All five sample files have a file size of 33280 bytes, buffer = 8192, first = 0, last = 8191, and therefore exactly match the expected size formula: $512 + 8192 * 4$.

The header contains fields for energy and FWHM calibration coefficients, but these fields are zero-filled in the supplied samples. The samples should therefore be imported as raw channel spectra unless calibration is provided externally.

The tSpectr structure contains DWORD *data. This pointer is an in-memory API member and is not stored as a pointer value in the file. The file stores the channel array after the 512-byte header.

Scope and inputs

Inputs reviewed: aspect.zip containing five .spc files, and the C header file named "Format of spectrum file.h". The header file is encoded as Windows-1251 / CP1251 and uses #pragma pack(1).

The comparison was made by reading the .h structure layout, deriving byte offsets, and validating those offsets against the actual binary values in each sample file.

Result

Result: MATCH. The documented tSpectrHeader layout corresponds to the observed binary layout in the sample files.

Confirmed file size formula

$\text{fileSize} = \text{sizeof}(\text{tSpectrHeader}) + \text{buffer} * \text{sizeof}(\text{DWORD}) = 512 + 8192 * 4 = 33280 \text{ bytes}$

Sample file validation matrix

File	Size	Expected size	buffer	first	last	Match	SHA-256 prefix
cs137 test2.spc	33280	33280	8192	0	8191	PASS	efec252d5b859247
co60 cs137 25m2.spc	33280	33280	8192	0	8191	PASS	e0c646ec3f9a2832
co60 test2.spc	33280	33280	8192	0	8191	PASS	4a85c382b4003c1c
cs137 test1.spc	33280	33280	8192	0	8191	PASS	d347c3f83ec2ea7d
cs137 test2 25mm2.spc	33280	33280	8192	0	8191	PASS	efec252d5b859247

Spectrum count statistics

File	Start time	liveTime raw	realTime raw	real-live delta	Total counts	Non-zero ch.	Max count @ ch.	Calibration fields
cs137 test2.spc	2016-01-13 21:00:13	7434877	7435634	757	51065	1269	1263 @ 74	zero
co60 cs137 25m2.spc	2016-01-14 19:55:04	4969813	4974207	4394	270090	2412	3898 @ 74	zero
co60 test2.spc	2016-01-10 23:44:19	4872964	4877246	4282	244126	1644	1591 @ 74	zero
cs137 test1.spc	2016-01-10 23:56:44	5618064	5623009	4945	281957	1742	1838 @ 77	zero
cs137 test2 25mm2.spc	2016-01-13 21:00:13	7434877	7435634	757	51065	1269	1263 @ 74	zero

Header field map and observed values

Offsets below are derived from `#pragma pack(1)` layout of `tSpectrHeader` and verified against the sample files.

Offset	Field	Type	Size	Meaning from .h	Observed in samples
0x0000	device[16]	BYTE[16]	16	Device name	All zero
0x0010	unit	BYTE	1	ADC number	0
0x0011	section	BYTE	1	Section number	0
0x0012	sampleTime[6]	BYTE[6]	6	Sample collection time, DMGHMS	All zero
0x0018	weight[16]	BYTE[16]	16	Mass, kg	All zero
0x0028	volume[16]	BYTE[16]	16	Volume, l	All zero
0x0038	startTime[6]	BYTE[6]	6	Acquisition start time, DMGHMS	Valid date/time in all samples
0x003E	liveTime	DWORD LE	4	Live time	Non-zero raw DWORD
0x0042	realTime	DWORD LE	4	Real time	Non-zero raw DWORD
0x0046	buffer	WORD LE	2	ADC buffer size / channel count	8192
0x0048	gain	WORD LE	2	ADC resolution/gain	0
0x004A	offset	WORD LE	2	ADC offset	0
0x004C	lowerLevel	WORD LE	2	Lower discriminator level	0
0x004E	upperLevel	WORD LE	2	Upper discriminator level	0
0x0050	first	WORD LE	2	First channel	0
0x0052	last	WORD LE	2	Last channel	8191
0x0054	chWidth	WORD LE	2	Channel width	0
0x0056	adcCont[10]	WORD[10] LE	20	Additional ADC parameters	All zero
0x006A	prepar	BYTE	1	Preparation type: 0-none, 1-ash	0
0x006B	preparTime[6]	BYTE[6]	6	Preparation time, DMGHMS	All zero
0x0071	reserve[15]	BYTE[15]	15	Reserved	All zero
0x0080	energy[4][16]	BYTE[64]	64	Energy calibration coefficients	All zero
0x00C0	fwhm[4][16]	BYTE[64]	64	FWHM calibration coefficients	All zero
0x0100	comment[4][64]	BYTE[256]	256	Measurement condition description	All zero
0x0200	data	DWORD[] LE	buffer*4	Spectrum counts, not a stored pointer	8192 uint32 values

Implementation notes and caveats

1. Read the header with explicit little-endian integer access or a packed structure on little-endian systems. Offset-based reading is safer for portable importers.
2. The date/time arrays use six raw numeric bytes in order: day, month, year since 2000, hour, minute, second. They are not text and not BCD in the supplied samples.
3. The liveTime and realTime fields are documented as DWORD values. The header does not define the unit or scale. Store them as raw values and derive duration only as an optional, empirical interpretation.

4. Do not confuse this Aspect binary SPC variant with Thermo/Galactic SPC, Ortec SPC, or text-header SpectraLine SPC variants.
5. Energy and FWHM calibration areas exist in the header, but in the supplied samples they are all zero. The importer should not invent keV calibration from these files.

Recommended importer sniffing rule

```
headerSize = 512
buffer = uint16_le(file[0x46])
first = uint16_le(file[0x50])
last = uint16_le(file[0x52])
canParse = buffer > 0 && first == 0 && last == buffer - 1
&& fileSize == 512 + buffer * 4
```

Conclusion

The header file is suitable as the primary specification for this Aspect .spc importer/exporter. The only unresolved item is the physical unit/scale of liveTime and realTime; the values should be preserved losslessly as raw DWORD fields.